DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) • CAPSTONE PROJECT SPECIFICATION

Customer Churn Intelligence & Retention System

Section 1: Project Planning & Management | Section 2: Literature Review | Section 3: Requirements Gathering

TARGET DATASET	BASELINE CHURN	PROJECT TIMELINE	TARGET RECALL
7,043 Rows	**26.54%**	**8 Weeks**	**≥ 80.0%**
IBM Telco (21 Features)	1,869 Churned Accounts	5 Agile Milestones	ROC-AUC ≥ 0.84

1. PROJECT PLANNING & MANAGEMENT

SECTION 01 // GOVERNANCE

1.1 Project Proposal — Overview, Objectives & Scope

Project Overview: Subscription-based businesses lose significant recurring revenue when customers cancel their subscriptions unexpectedly. Using the **IBM Telco Customer Churn dataset** (`blastchar/telco-customer-churn`, comprising 7,043 customer records and 21 service, billing, and demographic attributes with a 26.54% baseline churn rate), this project delivers an end-to-end **Customer Churn Intelligence & Retention System** that identifies churn patterns and predicts high-risk accounts before defection occurs.

Core Objectives: (1) Conduct SQL-based customer cohort and revenue-at-risk analysis; (2) Execute rigorous data cleaning (resolving missing `TotalCharges` strings), statistical hypothesis testing, and exploratory data analysis (EDA); (3) Build an interactive business intelligence dashboard for executive and retention teams; (4) Train, tune, and compare multiple classification models; and (5) Generate calibrated customer-level churn probability scores paired with actionable retention recommendations.

Project Scope: *In-Scope:* Relational PostgreSQL database design, Python EDA & feature engineering, training 5 supervised ML classifiers, SHAP explainability, and an interactive Power BI / Streamlit dashboard. *Out-of-Scope:* Live telecom network hardware telemetry and automated outbound billing mutation in third-party ERP systems.

1.2 Project Plan — Timeline (Gantt Chart), Milestones, Deliverables & Resource Allocation



Figure 1.1: 8-Week Project Execution Gantt Chart, Milestones (M1–M5), and Deliverable Gates (Resources: PostgreSQL 16, Python 3.11/Scikit-Learn, Power BI, GitHub)

1.3 Task Assignment & Roles — Defined Team Responsibilities

Role / Title	Assigned Member	Core Responsibilities & Owned Deliverables
Project Manager & Lead Data Analyst	`[Member 1 Name]`	Owns project schedule, stakeholder alignment, requirement tracking, statistical hypothesis testing (Chi-Square & Mann-Whitney U), and final presentation deck coordination.
Data Engineer & SQL Architect	`[Member 2 Name]`	Designs the 3NF relational schema (`dim_customers`, `dim_subscriptions`, `fact_billing_churn`), writes analytical SQL queries (CTEs, window functions), and resolves `TotalCharges` missingness.
Machine Learning Engineer	`[Member 3 Name]`	Builds the leak-free `ColumnTransformer` preprocessing pipeline, trains/compares 5 classification models (LogReg, RF, XGBoost, LightGBM, CatBoost), tunes threshold ($\tau^*=0.36$), and runs SHAP.

1.4 Risk Assessment & Mitigation Plan

Identified Risk Factor	Probability	Impact	Proactive Mitigation & Contingency Plan
Class Imbalance Bias (26.54% Churn vs 73.46% Stay)	HIGH	HIGH	Apply cost-sensitive class weights (`scale_pos_weight`), evaluate via PR-AUC and Recall rather than raw accuracy, and tune decision threshold ($\tau^* = 0.36$).
Data Leakage & Dirty Types (`TotalCharges` whitespace strings)	MEDIUM	HIGH	Coerce 11 blank strings where `tenure = 0` to `0.00` during ETL; encapsulate all scalers and encoders strictly inside cross-validation folds.
Black-Box Model Distrust (Business rejects ML scores)	MEDIUM	MEDIUM	Pair boosted tree probability outputs with local SHAP force values so every high-risk customer record displays its top 3 human-readable churn drivers.
Cross-Sectional Snapshot Limit (Lack of daily usage logs)	LOW	MEDIUM	Engineer longitudinal proxy ratios such as `Charge_Velocity_Ratio` and `Protection_Bundle_Score` to capture price hikes and service stickiness.

1.5 KPIs (Key Performance Indicators) — Project & System Success Metrics

KPI Category	Specific Performance Indicator	Target Benchmark	Measurement & Verification Method
Predictive ML Accuracy	Churn Recall (Sensitivity) & ROC-AUC	Recall ≥ 80% | AUC ≥ 0.84	Evaluated on 15% stratified holdout test set (1,057 customers).
System Performance	Dashboard Query & Filter Response Time	< 1.5 Seconds	Measured across interactive slicer transitions and watchlist filtering.
Reliability & Availability	Dashboard & Scoring Pipeline Uptime	≥ 99.5% Uptime	Automated nightly batch scoring SLA verification without ETL failure.
Business & User Adoption	Retention Team Adoption & Lift Capture	≥ 85% Adoption Rate	Top 2 risk tiers (38.8% of base) capture ≥ 80% of all actual churners.

2. LITERATURE REVIEW & EVALUATION FRAMEWORK

SECTION 02 // ACADEMIC REVIEW

Domain Synthesis & Prior Work: Academic literature on telecom and subscription churn prediction (e.g., Verbeke et al., 2012; Ahmad et al., 2019) demonstrates that ensemble gradient boosted decision trees (XGBoost, LightGBM, CatBoost) consistently outperform traditional logistic regression and standalone decision trees by capturing non-linear interactions between contract duration, billing friction, and value-added service bundles. Furthermore, recent studies emphasize *profit-driven threshold optimization* and *SHAP (SHapley Additive exPlanations)* to bridge the gap between raw probability estimation and actionable CRM intervention.

2.1 Feedback & Evaluation (Lecturer's Assessment) & 2.2 Suggested Improvements

- **Lecturer's Assessment & Feedback:** *"The project addresses a high-impact business problem using a well-validated industry dataset (IBM Telco Churn). Combining SQL warehouse querying, statistical EDA, multi-model benchmarking, and an interactive dashboard demonstrates strong end-to-end data science maturity. Ensure the team avoids evaluating models purely on accuracy due to the 26.5% class imbalance."* — **Status / Grade Check:** `[Approved — Exceeds Baseline Scope / Supervisor Sign-Off: _____]`
- **Suggested Improvements (Areas for Enhancement):**
  1. **Cost-Sensitive Thresholding:** Replace the default 0.50 cutoff with a business cost matrix comparing False Negative LTV loss ($500) vs. False Positive retention incentive cost ($65).
  2. **Probability Calibration:** Apply Platt scaling / Isotonic regression so predicted probabilities accurately reflect true empirical risk when multiplied by Monthly Charges.
  3. **Future Longitudinal Expansion:** In Phase 2, integrate time-series usage logs (e.g., *KKBox Churn Dataset*) or geospatial reason codes (*Maven Telecom Churn*) for survival hazard modeling.

2.3 Final Grading Criteria — Marks Breakdown Rubric (100 Points Total)

3. REQUIREMENTS GATHERING

3.1 Stakeholder Analysis — Key Stakeholders & Core Needs

Stakeholder Group	Organizational Role	Primary Information & System Needs	Primary Deliverable Used
VP of Customer Success & Execs	Executive Sponsor	Macro visibility into monthly churn rate, total MRR at risk ($139.1K/mo), and net annual revenue saved by interventions.	Dashboard Tab 1 (Executive KPI Pulse)
Retention & CRM Specialists	Primary Operational User	Prioritized daily watchlist of high-risk customers with individual churn probabilities, top 3 risk reasons, and next-best offer.	Dashboard Tab 3 (Customer Action Watchlist)
Product & Pricing Managers	Strategic Analyst	Cohort breakdown showing how contract terms, Fiber Optic pricing, and security/support bundles drive customer stickiness.	SQL Cohort Reports & Dashboard Tab 2
Data Science & Analytics Team	System Maintainers	Reproducible SQL schema, modular Python training pipeline, calibrated model comparison metrics, and drift monitoring.	GitHub Codebase & Model Registry

3.2 User Stories & Use Cases — System Interaction Scenarios

ID	Actor / Persona	User Story (As a... I want to... So that...)	Acceptance Criteria
US-01	Retention Specialist	*As a* Retention Agent, *I want to* filter active customers by **Critical Risk ($P \ge 0.75$)** and sort by Monthly Charges, *so that* I contact the highest-value flight risks first.	Table updates in <1.5s and exports target customer IDs to CSV.
US-02	Retention Specialist	*As a* Retention Agent, *I want to* view the top 3 SHAP risk factors for a specific customer (e.g., Month-to-Month, Tenure=2m, Electronic Check), *so that* I pitch the right offer.	Tooltip displays local feature contributions and prescribed offer.
US-03	Executive Sponsor	*As a* VP of Customer Success, *I want to* track Monthly Recurring Revenue (MRR) at risk across contract tiers and tenure bands, *so that* I allocate retention budgets effectively.	KPI cards dynamically recalculate MRR at risk across selected slicers.
US-04	Marketing Strategist	*As a* Product Marketer, *I want to* compare churn rates between customers with 0 vs. ≥ 2 protection add-ons (Tech Support, Online Security), *so that* I design sticky bundles.	SQL view and heatmap display exact churn rate lift per bundle score.
US-05	Data Scientist	*As a* Data Scientist, *I want to* benchmark 5 classifiers across ROC-AUC, PR-AUC, Recall, F1, and Brier Score, *so that* only the best-calibrated model enters production.	Automated evaluation script logs cross-validated metrics and confusion matrices.

3.3 Functional Requirements (FR)

- **FR-01 (SQL Cohort & Revenue Analysis):** The system shall store normalized customer, service, and billing tables in SQL and execute CTE/window queries calculating churn rate and lost MRR across contract, payment, and tenure segments.
- **FR-02 (Data Cleaning & Imputation):** The pipeline shall automatically detect and impute the 11 whitespace strings in `TotalCharges` (where `tenure = 0`), standardize binary flags, and collapse redundant `"No internet service"` labels.
- **FR-03 (Feature Engineering & Statistical EDA):** The system shall compute domain features

3.4 Non-Functional Requirements (NFR)

- **NFR-01 (Performance & Latency):** Batch scoring of all 7,043 customer records shall complete in under **10 seconds**; interactive dashboard slicer filtering and drill-downs shall render in under **1.5 seconds**.
- **NFR-02 (Scalability):** The modular SQL schema and `scikit-learn` preprocessing pipeline shall scale horizontally to support up to **1,000,000 subscriber records** without architectural refactoring.
- **NFR-03 (Security & Privacy Compliance):** All customer records shall be pseudonymized via alphanumeric `customerID` keys; zero PII (names, raw credit card numbers, or street addresses) shall be